

Implicit Differentiation and Inverse Functions

ANSWER KEY



Implicit differentiation basics

- Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.
 $2x + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{x}{y}$.
- Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.
 $3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$; $\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$, so $\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$.
- Find $\frac{dy}{dx}$ for $\sin(xy) = x$.
 $\cos(xy)(y + x\frac{dy}{dx}) = 1$: $y\cos(xy) + x\cos(xy)\frac{dy}{dx} = 1$, so $\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy)}$.

Tangent lines to implicit curves

- Find the equation of the tangent line to $x^2 + y^2 = 25$ at the point $(3, 4)$.
 $\frac{dy}{dx} = -\frac{x}{y}$; at $(3, 4)$: slope = $-\frac{3}{4}$. Tangent: $y - 4 = -\frac{3}{4}(x - 3)$, i.e. $y = -\frac{3}{4}x + \frac{25}{4}$.
- Find $\frac{dy}{dx}$ at the point $(1, 2)$ on the curve $x^2y + y^3 = 10$.
 $2xy + x^2\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$: $\frac{dy}{dx}(x^2 + 3y^2) = -2xy$, so $\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$. At $(1, 2)$: $\frac{-4}{1 + 12} = -\frac{4}{13}$.

Second derivatives via implicit differentiation

- Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 1$, expressing your answer in terms of x and y .
 $\frac{dy}{dx} = -\frac{x}{y}$. Differentiate again: $\frac{d^2y}{dx^2} = -\frac{y - x\frac{dy}{dx}}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{1}{y^3}$ (using $x^2 + y^2 = 1$).

Derivatives of inverse functions

- If $f(x) = x^3 + 2x + 1$, find $(f^{-1})'(4)$, using $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ and noting $f(1) = 4$.
 $f'(x) = 3x^2 + 2$; $f'(1) = 5$. $(f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{5}$.
- State the derivatives of the inverse trigonometric functions $\sin^{-1}x$, $\cos^{-1}x$, and $\tan^{-1}x$.
 $\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$; $\frac{d}{dx}\cos^{-1}x = -\frac{1}{\sqrt{1-x^2}}$; $\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$.
- Differentiate $f(x) = \tan^{-1}(3x)$.
 $f'(x) = \frac{3}{1+(3x)^2} = \frac{3}{1+9x^2}$.

Implicit differentiation with logarithmic and exponential terms

- 10 Find $\frac{dy}{dx}$ for $e^y = xy$.
 $e^y \frac{dy}{dx} = y + x \frac{dy}{dx} \cdot \frac{dy}{dx} (e^y - x) = y$, so $\frac{dy}{dx} = \frac{y}{e^y - x}$.
- 11 Find $\frac{dy}{dx}$ for $\ln(x + y) = x^2$.
 $\frac{1 + \frac{dy}{dx}}{x + y} = 2x$: $1 + \frac{dy}{dx} = 2x(x + y)$, so $\frac{dy}{dx} = 2x(x + y) - 1$.

Related applications of implicit differentiation

- 12 Find all points on the curve $x^2 + xy + y^2 = 7$ where the tangent line is horizontal.
Differentiate: $2x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$: $\frac{dy}{dx} = \frac{-2x - y}{x + 2y}$. Horizontal tangent needs $-2x - y = 0$, i.e. $y = -2x$.
Substituting into the curve: $x^2 + x(-2x) + (-2x)^2 = 7 \Rightarrow x^2 - 2x^2 + 4x^2 = 7 \Rightarrow 3x^2 = 7 \Rightarrow x = \pm \sqrt{7/3}$.
Points: $(\pm \sqrt{7/3}, \mp 2\sqrt{7/3})$.